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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

MTH 101: Elementary Mathematics I - Algebra, Trigonometry and Sequences

Close the secondary-school gaps that make first-year mathematics feel impossible

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Mathematics | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

MTH 101 assumes you already own certain algebraic habits. Many freshers do not, and the course moves fast. This pack rebuilds those habits - factorisation, indices, quadratic theory, trigonometric identities and sequences - with short explanations and worked examples that model good layout.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Factorise and simplify algebraic expressions including surds and partial fractions.
- Solve quadratic equations and use the sum and product of roots.
- Apply trigonometric ratios, identities and the sine and cosine rules.
- Work confidently with arithmetic and geometric progressions, including sums to n terms.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Algebraic fluency

Speed and accuracy in algebra decide your grade more than clever tricks do.

- Know the difference of two squares, perfect squares and the factor theorem.
- Rationalise surds by multiplying by the conjugate.
- Solve simultaneous equations by substitution when one equation is linear.
- For quadratics, use factorisation first, completing the square second and the formula last.
- For ax^2 + bx + c = 0, the sum of roots is -b/a and the product is c/a.
- Check answers by substitution - it takes seconds and catches sign errors.

> **Show your line of reasoning**
> Marks are awarded for correct intermediate steps. A bare answer that is wrong earns nothing; a clear method with a slip often earns most of the marks.

Trigonometry that stays consistent

Trigonometry rewards a diagram and punishes memorised ratios applied blindly.

- Use the unit circle to determine the sign of each ratio in each quadrant.
- Sine rule: a/sin A = b/sin B = c/sin C.
- Cosine rule: a^2 = b^2 + c^2 - 2bc cos A.
- Area of a triangle: half ab sin C.
- Core identities: sin^2 + cos^2 = 1; tan = sin/cos; double-angle formulae.
- Convert between degrees and radians before using calculus formulae.

Sequences and series

Identify the type of sequence before touching a formula.

- Arithmetic progression: common difference d; nth term a + (n-1)d; sum n/2(2a + (n-1)d).
- Geometric progression: common ratio r; nth term ar^(n-1); sum a(1 - r^n)/(1 - r) for r not equal to 1.
- Sum to infinity exists only when the absolute value of r is less than 1, and equals a/(1 - r).
- For word problems, write the first three terms explicitly to confirm the pattern.
- Binomial expansion: use Pascal's triangle for small powers, the general term for larger ones.

Sets, logic and proof

Set notation is the grammar of mathematics: union, intersection, complement, subset, cardinality. Venn diagrams solve many counting problems, and the inclusion-exclusion principle generalises them: for two sets, the size of the union is the sum of the sizes minus the overlap.

Logic underlies proof. A statement and its contrapositive are equivalent; the converse and inverse are not. Proof by contradiction assumes the negation and derives a contradiction. Proof by induction establishes a base case and an inductive step, and is valid only over the natural numbers.

The commonest student error is proving an example rather than a general claim. Three cases support a conjecture but do not prove it; one counter-example refutes it.

Algebra, functions and equations

Quadratic equations are solved by factorisation, completing the square, or the formula. The discriminant determines the nature of the roots: positive for two real roots, zero for a repeated root, negative for complex roots. The sum and product of roots follow directly from the coefficients.